UNIVERSITY OF PERADENIYA
Faculty of Engineering

TUTORIAL ON BODE DIAGRAMS

EE 352 AUTOMATIC CONTROL

1. The transfer function of a plant is given by

$$G_4(s) = \frac{100}{(s+1)(s+2)}$$

a) Calculate the phase margin and the gain crossover frequency of the plant. **(40% marks)**

b) The Bode diagram of the plant is shown in Figure 1. Use the Bode diagram to determine the following:

   i. phase margin **(15% marks)**

   ii. gain crossover frequency **(15% marks)**

   iii. additional gain required to make the phase margin 45^0 **(30% marks)**

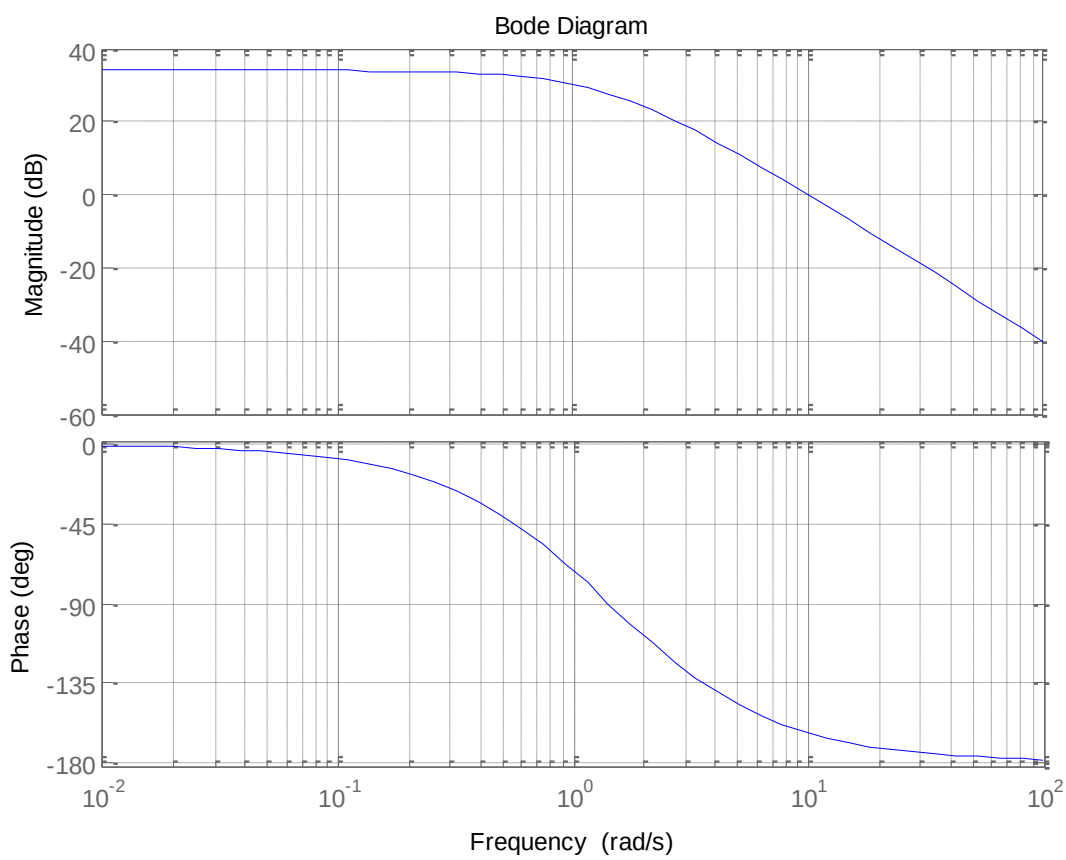

Figure 1

2. Consider the systems with the transfer functions given below:

   a) $G(s) = \frac{1}{s(s+2)(s+4)}$

   b) $G(s) = \frac{1}{(s+1)(s+2-j)(s+2+j)}$

   c) $G(s) = \frac{(s+2)}{(s+1)(s+3-j)(s+3+j)}$

   d) $G(s) = \frac{1}{s(s+1)(s+3)(s+4)}$

i. Find the break frequencies and sketch the asymptotic Bode diagrams of the systems.

ii. Use the asymptotic Bode diagrams of the systems to draw the Bode diagrams of the same systems.